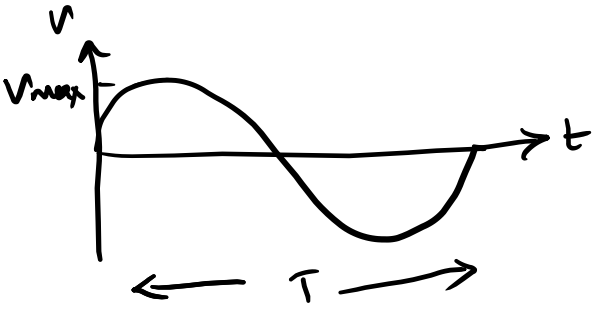
AC Circuits

$V = V_{max} \sin \omega t$

V: Instantaneous voltage.

V_{max}: Max voltage (or) amplitude.

ω: angular freq.

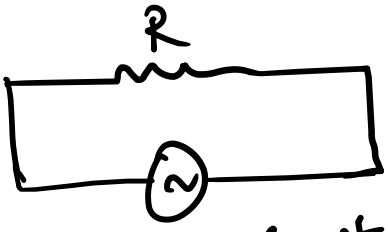


I is also sinusoidal.

I can be in phase with 'V' (or)

I can be out of phase with 'V' (lead / lag)

Resistors in AC:



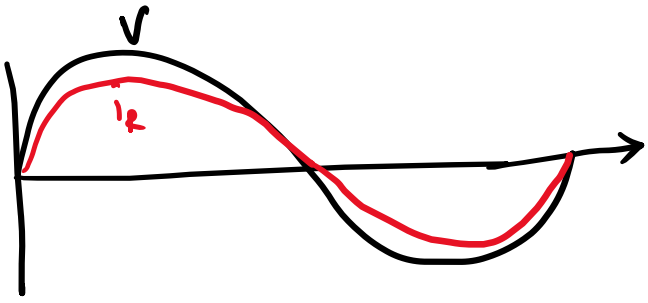
$V = V_{max} \sin \omega t$.

instantaneous current.

$$i_R = \frac{V_R}{R}$$

$$V_R = I_{max} \cdot R \cdot \sin \omega t$$

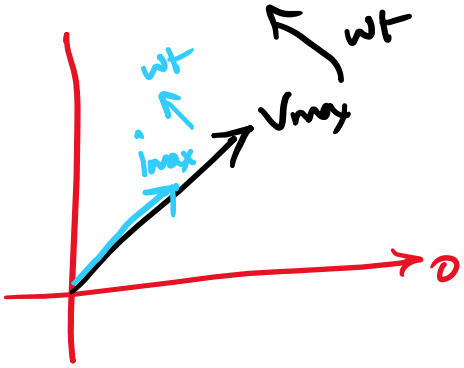
$$i_R = I_{max} \cdot \sin \omega t.$$



Current through a resistor will be in phase with applied sinusoidal voltage.

Phase diagram

Vector along with phase angle.



rms voltage and current

$$I_{rms} = \frac{\hat{i}_{max}}{\sqrt{2}} = 0.707\,\hat{i}_{max}$$

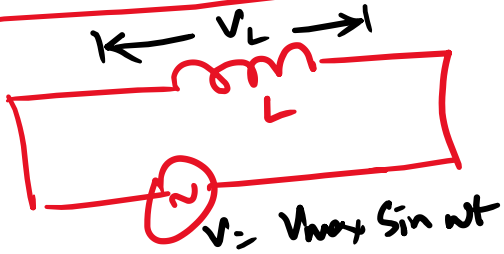
$$V_{rms} = \frac{V_{max}}{\sqrt{2}} = 0.707\,V_{max}$$

$P = \hat{i}^2 \cdot R$ ($\hat{i}$: instantaneous value.)

I_{max} only occurs for an instant.

$$\boxed{P_{avg} = i_{rms}^2 \cdot R} \simeq i_{DC}^2 \cdot R$$

Inductor in an AC circuit:



$$V + V_L = 0$$

$$V_L = -L \frac{di_L}{dt}$$

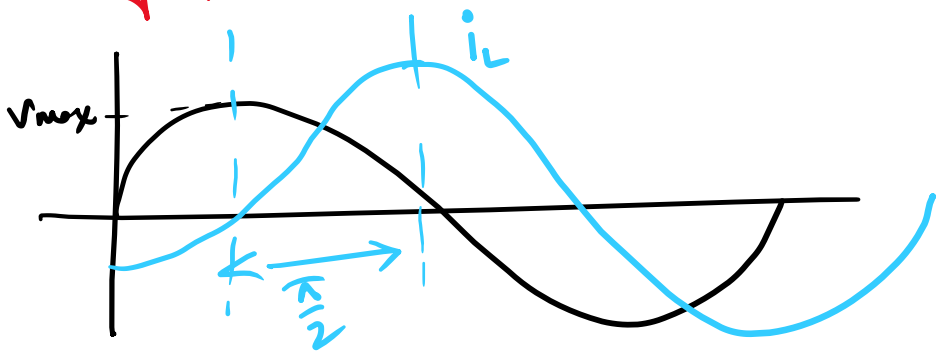
$$V = L \frac{di_L}{dt}$$

$$i_L = \int \frac{V_{max}}{L} \sin \omega t \, dt$$

$$i_L = -\frac{V_{max}}{\omega L} \cos \omega t$$

$$\hat{i}_L = \frac{V_{max}}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$$

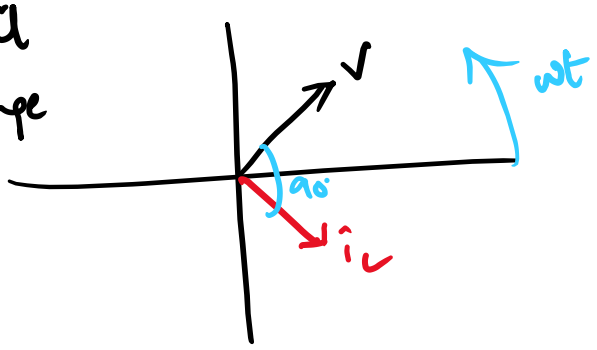
i_L is out of phase with V by $\frac{\pi}{2}$



i reaches peak later.

i_L lagging V

Purely inductive circuit
→ current lags voltage by 90°



Inductive Reactance:

ωL has the same unit as 'R'.

$X_L = \omega L$ = Inductive Reactance.

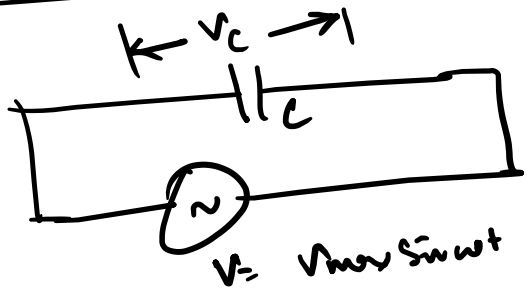
$I_{max} = \frac{V_{max}}{X_L}$; $I_{rms} = \frac{V_{rms}}{X_L}$

Voltage across inductor : $V_L = -L\frac{di}{dt}$

$= -V_{max}\sin\omega t$

$= -I_{max} X_L \sin\omega t$.

Capacitor in AC circuit

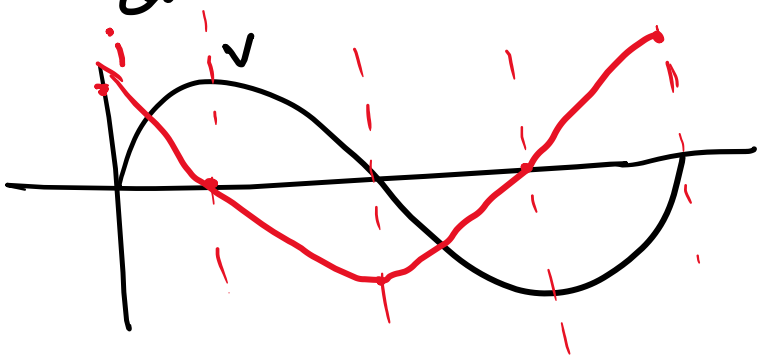


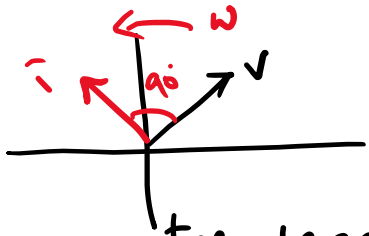
$V_C = V = V_{max}\sin\omega t$

$i = \frac{dq}{dt}$

$q = CV \Rightarrow q = C V_{max}\sin\omega t$

$i_C = \frac{dq}{dt} = \omega C V_{max}\sin(\omega t + \frac{\pi}{2})$





Current through a capacitor leads the applied voltage by 90°

Capacitive Reactance:

$$I_{max} = \omega C \cdot V_{max} = \frac{V_{max}}{1/\omega C}$$

$$X_C = \frac{1}{\omega C}$$